

Chemistry Ch 2.5: Colloids and Suspensions

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Pichhle safhay par solution tha. Ab do aur mixtures. Teenon ka farq sirf **ek cheez** se hai: **zarray ka size**. Baqi sab usi se nikalta hai.

Zarray ka size		Pehchan aur misalein
Solution	1 nm se kam	Homogeneous, mustahkam. Namak ka paani.
Colloid	1 se 1000 nm	Heterogeneous, Tyndall effect dikhata hai, aur alag nahin hota . Doodh, nishasta, khoon, saabun.
Suspension	1000 nm se bara	Zarray nazar aate hain, aur rakha rahe to neeche baith jate hain. Chalk ka paani, paint, gaara.

Tyndall effect, aur ise pehchanne ka tareeqa

Jab roshni ka shuaa colloid mein se guzre to **raasta nazar aa jata hai**, kyunke zarray roshni ko bikher dete hain.

Aap ne ye dekha hua hai: andheray kamre mein khirki se aane wali dhoop ki lakeer, ya raat mein dhund mein gaari ki headlight. Solution mein aisa nahin hota, kyunke us ke zarray bohat chhote hain.

Table 2.3 ke chaar farq, tarteeb se

- 1. Particle size:** chhota, darmiyana, bara.
- 2. Filtration:** solution aur colloid **filter paper se guzar** jate hain, suspension **ruk** jata hai.
- 3. Light scattering:** sirf **colloid** Tyndall effect dikhata hai.
- 4. Stability:** solution mustahkam, colloid mustahkam, suspension **baith** jata hai.

Quiz

1. Give the particle size range for a **solution**, a **colloid** and a **suspension**, with one example of each.
2. What is the **Tyndall effect**, and which of the three shows it? Then give two more differences between a colloid and a suspension.